

Spec: Future Leave Accrual Projection

Status

Not implemented

Problem

When an employee books leave in advance, the system shows their **current** available balance rather than what they will have accrued by the time the leave starts. This produces false insufficient-balance warnings for future leave, and gives employees and HR no way to verify forward-looking coverage.

Example

- Today (April 2026): Employee has 2.5 days annual leave available.
- They apply for 5 days starting August 2026.
- They accrue 1.25 days/month x 4 months = 5 more days by August.
- After passing through a tenure tier at month 24 they may accrue at a higher rate.
- The system currently warns them they are 2.5 days short -- but they will be covered.

Scope

- Projected available balance at the leave start date, shown on the **leave request form** once dates are selected.
- The projection must be **exact** given available data: it processes each future month individually using the same logic the real accrual engine uses.
- Handles annual leave cycle resets (carry-over rollover) when the projection window crosses a cycle boundary.
- No changes to actual balance records -- display/advisory only.
- **Sick leave is excluded** -- sick leave is granted in arrears and cannot be projected meaningfully. Show current balance only for sick leave.
- **Leave applications are capped at 12 months in the future** (see below). This guarantees the projection window crosses at most one cycle boundary.

12-Month Application Limit

Leave requests with a start date more than 12 months from today must be rejected at both the backend (**POST /api/leave-requests**) and the frontend (disable date selection beyond the limit on the calendar).

Why this matters for the projection: the annual leave cycle is 12 months. Capping at 12 months ensures the projection window can cross at most one cycle boundary. Projecting through two resets is not handled and not needed.

Algorithm

The projection mirrors the monthly accrual loop in **server/index.ts** exactly. It maintains a running simulation of **projected_total**, **projected_carry_over**, and **projected_carry_over_expiry**.

Initialise:

```
projected_total          = current_balance.total
projected_carry_over     = current_balance.carryOverDays
projected_carry_over_expiry = current_balance.carryOverExpiry
```

For each future month M from (last_accrued_month + 1) to (month containing asOfDate - 1):

```
1. completed_months = completedMonths(user.startDate, last_day_of_M)
   -> Rate tier can change mid-projection when employee crosses a tier threshold.

2. { rate } = determineAccrualRate(completed_months, user.annualLeaveOverrideDays, tiers)

3. pausing_days = calendar days of approved accrual-pausing leave
                  (Unpaid, Maternity, Parental, Adoption, Commissioning)
                  falling within month M, from actual approved request records.

4. { activeDays, totalDays } = calculateActiveDays(M.year, M.month, user.startDate, null, pausing_days)

5. accrual_M = calculateAnnualLeaveAccrual(rate, activeDays, totalDays)

6. projected_total += accrual_M

7. CHECK: is M the cycle-end month? (the month before annual_leave_cycle_start)
   If YES -> simulate the annual cycle rollover (see below).

8. Record breakdown entry for M.
```

Annual cycle rollover simulation (step 7):

When the loop reaches the cycle-end month, before moving to the next month:

```
// What would be available at the end of this month?
pending_post_reset = sum of working days of approved requests
                    whose start date falls AFTER the cycle end
                    (pre-reset pending requests settle before the reset runs)

available_at_reset = projected_total - current_taken - pending_post_reset

// Move available balance to carry-over
new_carry_over      = max(0, available_at_reset)
new_carry_over_expiry = cycle_end_date + graceMonths (from leave_carry_over_grace_months setting)

// Zero the running total and restart
projected_total      = 0
projected_carry_over  = new_carry_over
projected_carry_over_expiry = new_carry_over_expiry

// Record in breakdown:
{ month: M, event: "cycle_reset", carryOverCreated: new_carry_over,
  carryOverExpiry: new_carry_over_expiry, accrual: accrual_M }
```

After the loop -- compute final projected available:

```
// Is current carry-over still valid at asOfDate?
carry_over_at_date = 0
if projected_carry_over > 0:
    if projected_carry_over_expiry IS NULL or projected_carry_over_expiry >= asOfDate:
        carry_over_at_date = projected_carry_over
    else:
        carry_over_at_date = 0 // expired before leave starts

// Pending at asOfDate = only requests dated on or after asOfDate
// (requests before asOfDate will have settled to taken by then)
pending_at_date = sum of working days of active requests
                    whose start date >= asOfDate

projected_available = projected_total
                    + carry_over_at_date
                    - current_taken
                    - pending_at_date
```

Worked Example: December -> February (Cycle Resets January 1)

Setup:

- Today: 1 December 2026
- Leave start date: 10 February 2027
- Current balance: **total = 15, taken = 5, pending = 0** -> available = 10 days
- Monthly accrual rate: 1.25 days/month
- No approved pausing leave
- Grace months: 6

Loop:

Month	Accrual	Event	Running total	Carry-over
Dec 2026	+1.25	Cycle reset at end of Dec-> 0		11.25 (expires 30 Jun 2027)
Jan 2027	+1.25	--	1.25	11.25

At February 10 2027:

- **projected_total** = 1.25
- **carry_over_at_date** = 11.25 (expires June 2027 -- still valid in February)
- **current_taken** = 5
- **pending_at_date** = 0

```
projected_available = 1.25 + 11.25 - 5 - 0 = 7.5 days
```

The employee has 7.5 days available in February -- the carry-over from December is **included**, not deducted. This is correct: those days carried over because the employee had earned them but not used them.

Note on December accrual timing: December's accrual (1.25 days) runs on 1 January. The rollover also happens in the December cycle-end processing. The projection adds the December accrual first, then applies the rollover, giving **total = 11.25** before the reset zeroes **total** and moves it to carry-over.

Data Requirements

Data	Source
Employee startDate, annualLeaveOverrideDays	users table
All accrual rate tiers	accrual_rate_tiers table
Current leave balance (total, taken, pending, carryOverDays, carryOverExpiry)	leave_balances table
All active leave requests for the user	leave_requests table -- needed for pausing leave and pending-at-date calculation
annual_leave_cycle_start setting	settings table -- format MM-DD, default 01-01
leave_carry_over_grace_months setting	settings table -- default 6
Last accrual run date	settings table key last_accrual_run, or derive as first day of current month

All available without schema changes.

Backend: New Endpoint

``GET /api/leave-balances/:userId/projected``

Query parameters:

- `leaveType` -- reject `Sick Leave` with 400.
- `asOfDate` -- `yyyy-MM-dd` -- must be a future date and within 12 months.

Response:

```
{
  "currentAvailable": 10.0,
  "projectedAvailable": 7.5,
  "cycleResetOccurs": true,
  "carryOverCreated": 11.25,
  "carryOverExpiry": "2027-06-30",
  "monthsProjected": 2,
  "breakdown": [
    {
      "month": "2026-12",
      "accrual": 1.25,
      "rate": 1.25,
      "activeDays": 31,
      "totalDays": 31,
      "tierNote": "Tier 1 (15 days/yr)",
      "event": "cycle_reset",
      "carryOverCreated": 11.25,
      "carryOverExpiry": "2027-06-30"
    },
    {
      "month": "2027-01",
      "accrual": 1.25,
      "rate": 1.25,
      "activeDays": 31,
      "totalDays": 31,
      "tierNote": "Tier 1 (15 days/yr)"
    }
  ]
}
```

Error responses:

- 400 if `leaveType = 'Sick Leave'`

- 400 if **asOfDate** is not a valid future date
- 400 if **asOfDate** is more than 12 months from today
- 400 if employee has no **startDate**

Implementation: **server/routes.ts** alongside the existing balance routes. Imports the same functions from **bcea.ts** already used by the accrual engine. Extract the core projection loop into a standalone function so it can also be called from **POST /api/leave-requests** for backend validation.

Frontend: Leave Request Form (`LeaveRequest.tsx`)

Date picker limit

Disable all dates more than 12 months from today on the calendar picker. Add a clear label: "Leave can be booked up to 12 months in advance."

When to show the projection

Both conditions must be true:

- Leave type is not Sick Leave.
- Leave start date is **more than 30 days** from today.

UI -- projected balance panel

Normal case (no cycle reset):

```

????????????????????????????????????????????????????????
? Projected balance at 1 Aug 2026                                ?
? Current available      2.50 days                               ?
? Accrual May-Jul       +3.75 days  (4 months x 1.25)           ?
? Projected available   6.25 days  v                             ?
? This request (5 days) will be covered.                        ?
????????????????????????????????????????????????????????

```

With a cycle reset:

```

????????????????????????????????????????????????????????
? Projected balance at 10 Feb 2027                                ?
? Current available     10.0 days                               ?
? Dec accrual           +1.25 days                               ?
? Annual cycle reset   -> 11.25 days carried over               ?
?   (expires 30 Jun 2027)                                         ?
? Jan accrual           +1.25 days  (new cycle)                 ?
? Projected available   7.5 days  v                             ?
? This request (5 days) will be covered.                        ?
????????????????????????????????????????????????????????

```

If projected is negative:

```

? Projected available  -1.25 days  x                               ?
? Your balance will be insufficient at the start date.?
? You can still submit -- HR will review.                      ?

```

Backend Validation Change

At submission (**POST** `/api/leave-requests`):

- If **asOfDate** > 12 months from today -> **reject** with a clear error.
- If current available < requested days **but** projected available >= requested days -> allow submission, add to **adminNotes**:

[Balance note: Current available insufficient (-0.5 days) but projected balance at 2027-02-10 is +7.5 days -

- If both current and projected are insufficient -> allow submission (HR discretion), add:

[Balance warning: Insufficient leave. Current: -0.5 days. Projected at 2027-02-10: -1.25 days. HR discretion

API Client (`client/src/lib/api.ts`)

```
projected: (userId: string, leaveType: string, asOfDate: string) =>
  apiRequest('GET', `/api/leave-balances/${userId}/projected?leaveType=${encodeURIComponent(leaveType)}&asOfDate=${asOfDate}`)
```

Edge Cases

Scenario	Handling
Sick leave	400 -- show current balance only
Leave starts <= 30 days away	Don't fetch projection -- show current balance only
Leave start > 12 months away	Rejected at backend and blocked on frontend calendar
No months to project	Return projectedAccrual: 0, show current balance
Cycle reset within projection window	Simulate rollover mid-loop; carry-over included in projected available
Carry-over from reset expires before leave start date	Excluded from projected available; noted in UI (only possible if grace period <
Current carry-over expires before leave start date	Excluded from projected_total calculation
Rate tier boundary crossed mid-projection	Each month uses the correct tier; tier change shown in breakdown
Employee has approved Unpaid Leave in a future month	Accrual for that month is pro-rated or zero
Employee has approved Unpaid Leave spanning the cycle-end	Carry-over created is reduced accordingly
annualLeaveOverrideDays set	determineAccrualRate() uses the override automatically
Last accrual run setting not found	Default to first day of current calendar month